

2.1 Functions of Several Variables and Graphing

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In this course we are interested in functions of the type

$$f: \underset{\text{domain}}{X \subseteq \mathbb{R}^n} \rightarrow \underset{\text{codomain}}{Y \subseteq \mathbb{R}^m}$$

$n = \#$ of independent variables
 $m = \#$ of components of f

e.g.: We have been talking a lot about two types of functions when discussing geometry

lines $l: \mathbb{R} \rightarrow \mathbb{R}^3$ or $l: \mathbb{R} \rightarrow \mathbb{R}^m$ $l(t)$ 1 independent var.
2 components

planes $\pi: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ $\pi(s, t)$ 2 independent vars.
3 components

e.g.: You can represent some area of land X in a map and make a function f for the altitude at each point

$f: X \subseteq \mathbb{R}^2 \rightarrow \mathbb{R}$ 2 ind. vars 1 comp.

def: A function $f: X \rightarrow Y$ is called one-to-one (or injective) if no two elements in the domain have the same image (i.e. if $x_1, x_2 \in X$ and $x_1 \neq x_2$, then $f(x_1) \neq f(x_2)$).

def: $\text{Range}(f) = \{y \in Y: y = f(x) \text{ for some } x \in X\}$ (another notation is $f(X) = \text{Range}(f)$).

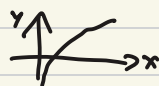
def: A function $f: X \rightarrow Y$ is called onto (or surjective) if every element in Y is the image of an element in X (i.e. $\text{Range}(f) = Y$).

e.g. Is this function 1-1 (and) on onto?

$$\ln(x) : (0, \infty) \rightarrow \mathbb{R} \quad 1-1 \text{ and onto}$$

$$f(x, y) = x^2 + y^2 : \mathbb{R}^2 \rightarrow \mathbb{R} \quad \text{not 1-1 not onto}$$

$$f(x, y) = x^2 + y : \mathbb{R}^2 \rightarrow \mathbb{R} \quad \text{not 1-1 is onto}$$

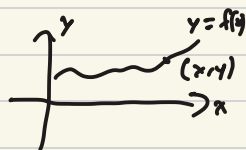


e.g. What is the largest domain $\subseteq \mathbb{R}^2$ for this function $\{(x, y) \mid x + y > 0\}$
 $\cdot \ln(x^2 + y^2)$ domain = $\mathbb{R}^2 \setminus \{(0, 0)\}$ $\cdot \ln(x + y)$ domain = $\{(x, y) \mid y > -x\}$

Graphing

We are more familiar with graphs of $f: X \subseteq \mathbb{R} \rightarrow \mathbb{R}$

This is a form of an explicit graph, but we will need more types of graphs:



1) Explicit graph: The domain and codomain are represented in different axes of the graph
 (mathematically this graph is $\{(\bar{x}, f(\bar{x})) : \bar{x} \in X\} \subseteq \mathbb{R}^n \times \mathbb{R}^m = \mathbb{R}^{n+m}$)

e.g. $f(x) = \sqrt{x}$
 $f: \mathbb{R}_+ \subseteq \mathbb{R} \rightarrow \mathbb{R}$

$$f(x) = (\cos x, \sin x)$$

$$f: \mathbb{R} \rightarrow \mathbb{R}^2$$

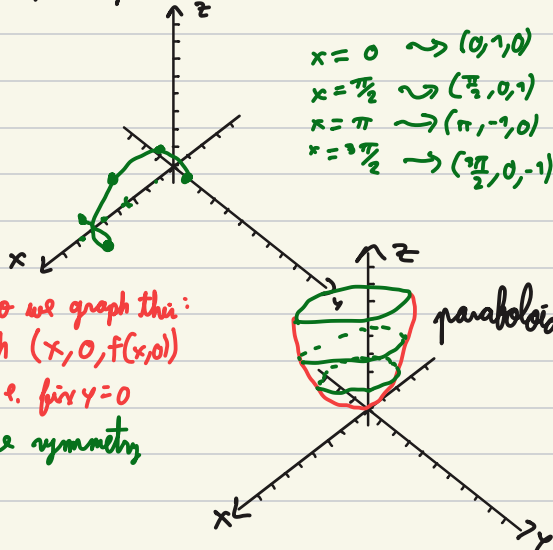
$$\text{graph} = \{(x, \cos x, \sin x) \mid x \in \mathbb{R}\}$$

$$f(x, y) = x^2 + y^2$$

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

how do we graph this:
 • let's graph $(x, 0, f(x, 0))$
 i.e. fix $y = 0$

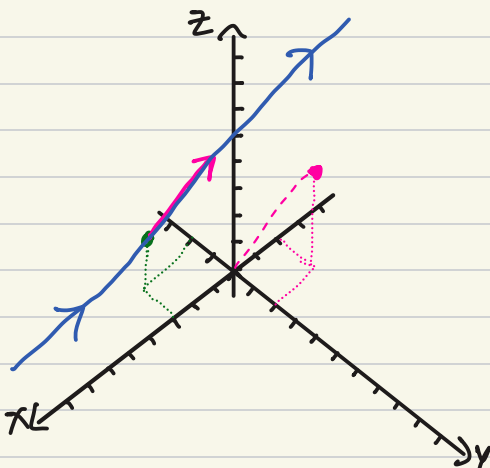
• then use symmetry



2) Parametric graphs: The axes of the domain variables are suppressed informally speaking, we are only interested in the image of the function and what shape it traces.

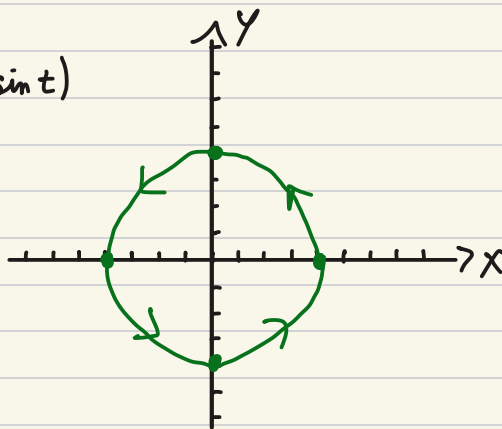
e.g.: $(x, y, z) = \vec{r}(t) = (3, -2, 2) + t(-2, 2, 3)$

$$\vec{r}: \mathbb{R} \rightarrow \mathbb{R}^3$$



e.g.: $(x, y) = r(t) = 4(\cos t, \sin t)$

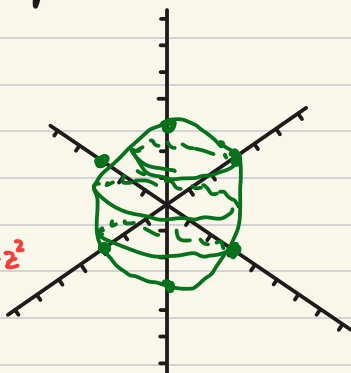
$$r: \mathbb{R} \rightarrow \mathbb{R}^2$$



- implicit graphs: The axes of the codomain are suppressed
 - Another way to think about it is we are graphing the points in the domain that satisfy certain equations.

e.g. graph $x^2 + y^2 + z^2 = 3$

It is convenient to think of this graph as depicting the level set of the function $f(x, y, z) = x^2 + y^2 + z^2$



Graphing surfaces inside $\mathbb{R}^3$

It is not easy to visualize how z change as x and y change and draw a 2-D (at scale and with perspective) representation of an object inside $\mathbb{R}^3$, so we break it down and investigate how (x, y) looks at different values of z .

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}$$

def: The contour curve at height c of f is the curve in $\mathbb{R}^3$ defined by $f(x, y) = c$ i.e. $\{(x, y, z) \mid z = f(x, y) \text{ and } z = c\}$

$$* z = f(x, y) = \frac{1}{12} y^3 - y - \frac{1}{4} x^2 + \frac{7}{2}$$

Figure 2.11 Some contour curves of the function in Example 10.

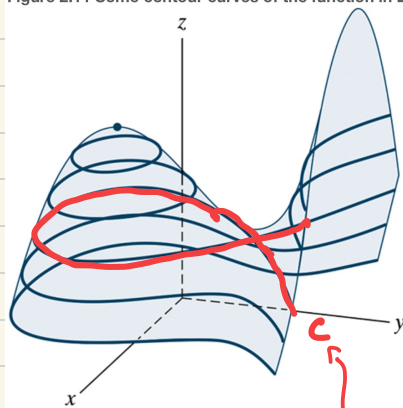
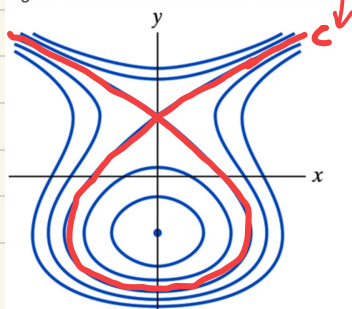


Figure 2.12 Some level curves of the function in Example 10.



def: The level curve at height c of f is the curve in $\mathbb{R}^2$ defined by $f(x, y) = c$ i.e. $\{(x, y) \mid f(x, y) = c\}$

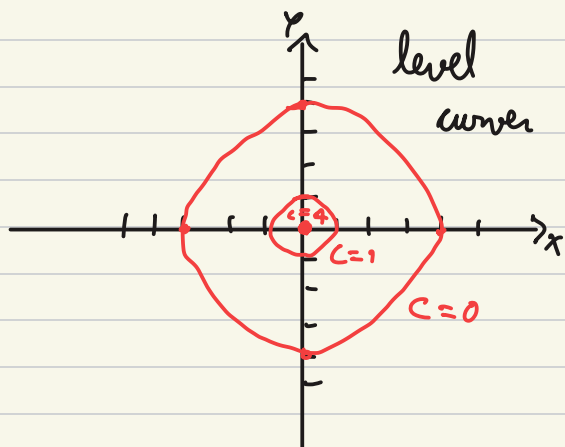
e.g. (section 2.1 example 11)

$$f(x,y) = 4 - x^2 - y^2$$

$$c = 4 \leadsto (x,y) = (0,0)$$

$$c = 3 \leadsto x^2 + y^2 = 1$$

$$c = 0 \leadsto x^2 + y^2 = 4$$



contour curves
and filling in

